COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	13/08/26	**DATE SUBMITTED:**	13/08/26
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Prelim Exam Skills **Prelim Exam Skills**

1. PROCEDURE OUTPUT

1. Set up

Replace the sample IP addresses below with the actual IP addresses of your virtual machines.

Hostname	Operating System	IP Address
server-1	ubuntu	192.168.56.111
server-2	ubuntu	192.168.56.112
centos1 CentOS	centos1 CentOS	

Part A: Installing and Verifying Ansible

Submit a screenshot of the following:

- Ansible version

```
localmachine@Localmachine-Andoy:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/localmachine/.ansible/plugins/modules'
, '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
localmachine@Localmachine-Andoy:~$ s
```

- SSH connectivity

Server-1

```
localmachine@Localmachine-Andoy:~$ ssh server-1@192.168.56.111
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

*** System restart required ***
Last login: Thu Jul 30 09:31:35 2026 from 192.168.56.110
server-1@server-1-andoy:~$
```

Server-2

```
localmachine@Localmachine-Andoy:~$ ssh serverr-2@192.168.56.112
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

*** System restart required ***
Last login: Thu Jul 30 09:31:51 2026 from 192.168.56.110
serverr-2@server-2-andoy:~$
```

Part B: Building an Inventory File

Demonstrate the following:

- inventory.yaml

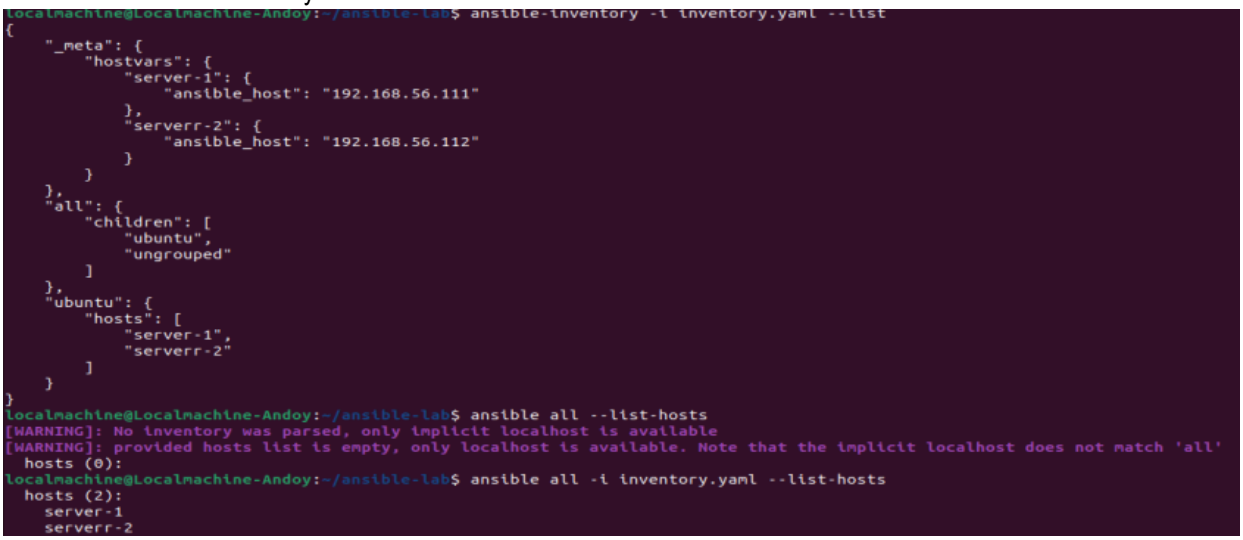
inventory.yaml

```yaml
all:
  children:

    ubuntu:
      hosts:
        server-1:
          ansible_host: 192.168.56.111
          ansible_user: server-1

        serverr-2:
          ansible_host: 192.168.56.112
          ansible_user: serverr-2
```

- Successful inventory verification

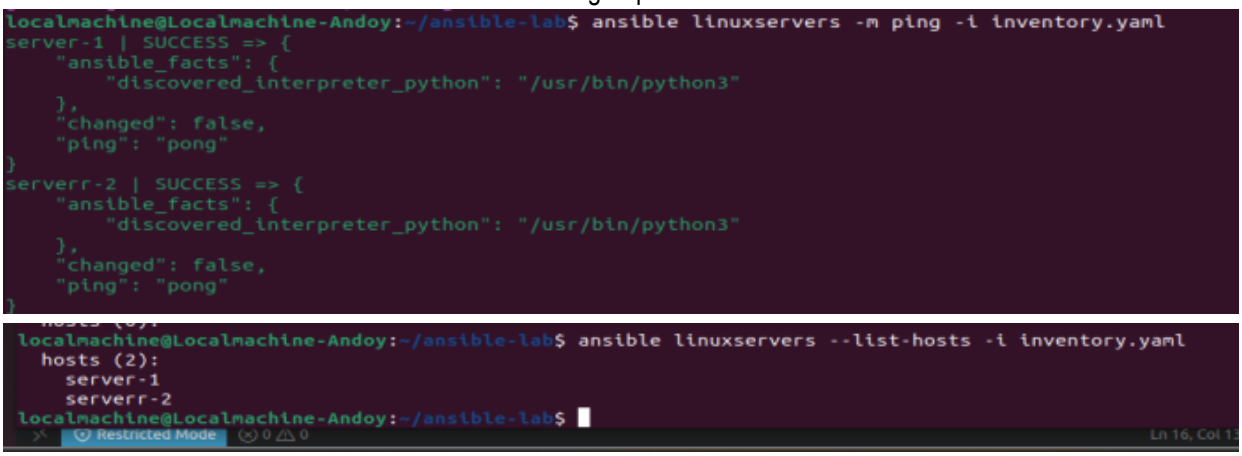


- Successful ping results



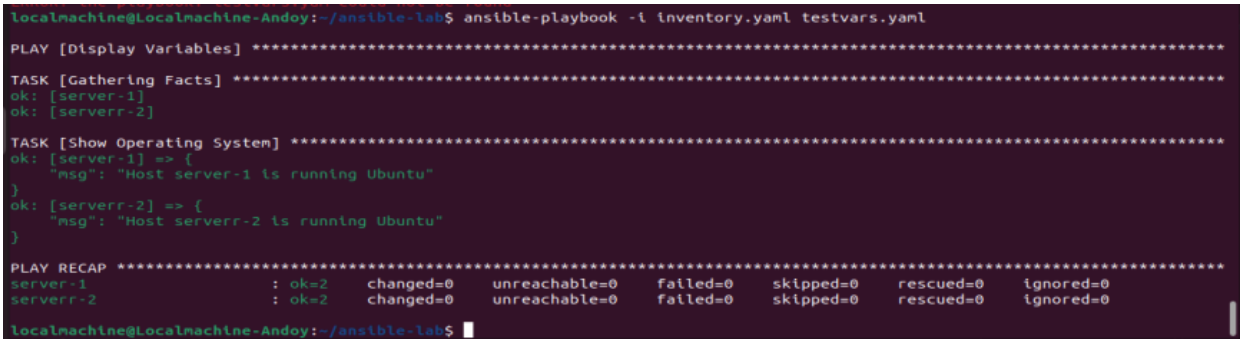
Part C: Creating a Metagroup

- Demonstrate successful execution of the metagroup.



Part D: Creating Variables

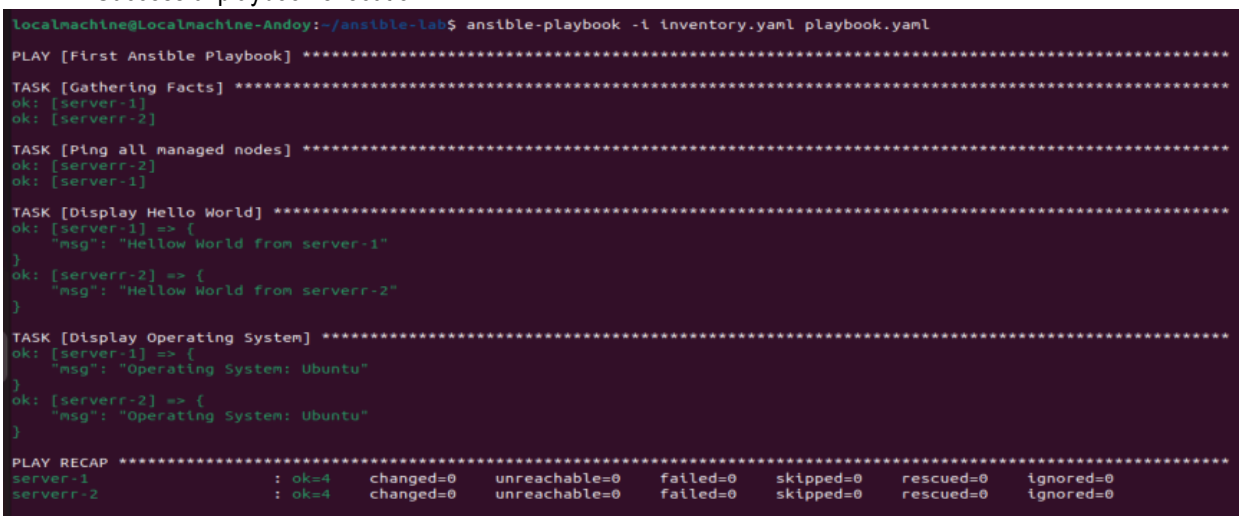
- Demonstrate successful use of variables.



Part E: Creating Your First Playbook

Demonstrate:

- Successful playbook execution



- Successful task completion on all managed nodes

```
inventory.yaml   testvars.yaml   playbook.yaml
playbook.yaml
1  ---
2  - name: First Ansible Playbook
3    hosts: linuxservers
4
5    tasks:
6      - name: Ping all managed nodes
7        ping:
8
9      - name: Display Hello World
10       debug:
11         msg: "Hellow World from {{ inventory_hostname }}"
12
13     - name: Display Operating System
14       debug:
15         msg: "Operating System: {{ os_type }}"
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Create a playbook named challenge.yaml that displays the message:

```
localmachine@Localmachine-Andoy:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [HAHAHHAHAHHAHAHA] ********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [server-1]
ok: [serverr-2]

TASK [Hello to all type shi] ***************************************************************************************************
ok: [server-1] => {
    "msg": "Welcome to Ansible Laboratory server-1"
}
ok: [serverr-2] => {
    "msg": "Welcome to Ansible Laboratory serverr-2"
}

PLAY RECAP *********************************************************************************************************************
server-1                   : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
serverr-2                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

localmachine@Localmachine-Andoy:~/ansible-lab$
```

Guide Questions

1. What is the role of the Control Node in Ansible?
   - Control Node is the central brain. This is like the cpu of a computer but an Ansible architecture where all automation commands and playbooks are executed.
2. Why is an inventory file required?
   - Because the inventory file is where the servers live and the information about the system. This stores the variables that have the information.
3. What is the purpose of a metagroup?
   - This metagroup contains the children rather than containing the individual hosts. This makes the grouping of variables easily.
4. How do variables simplify Ansible administration?
   - This enables us to use the variables dynamically and I can reuse them everywhere.
5. What is the difference between an ad hoc command and a playbook?
   - An ad hoc command is a single, quick command line instruction used to perform a simple task like the ping

of servers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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